

1. (Revision on first-year calculus)

- (a) Find the positive number(s) x for which the distance between the points $A(1, -1, x)$ and $B(-2x, 3, 1)$ is equal to $\sqrt{19}$.

We seek positive number(s) x such that

$$|\overrightarrow{AB}| = \sqrt{(-2x-1)^2 + (3+1)^2 + (1-x)^2} = \sqrt{19} \Leftrightarrow \boxed{5x^2 + 2x - 1 = 0}.$$

and the only positive solution of the quadratic equation is $x = \frac{-1 + \sqrt{6}}{5}$.

- (b) Find the cartesian equation of the plane Q through the point $(1, -1, 3)$ and parallel to the plane with cartesian equation $3x + y + z = 7$.

The plane Q has the same normal as the plane with cartesian equation $3x + y + z = 7$. That is, the plane Q has normal vector $(3, 1, 1)$. Hence, the plane Q has cartesian equation

$$\begin{aligned} (x-1, y+1, z-3) \cdot (3, 1, 1) &= 0 \Leftrightarrow 3 \times (x-1) + 1 \times (y+1) + 1 \times (z-3) = 0 \\ &\Leftrightarrow \boxed{3x + y + z = 5}. \end{aligned}$$

- (c) Find an equation satisfied by the coordinates of the point $P(x, y, z)$ lying at equal distance from the points $A(-1, 3, 2)$ and $B(1, 0, 4)$.

$$\begin{aligned} |\overrightarrow{AP}| &= |\overrightarrow{BP}| \Leftrightarrow \sqrt{(x+1)^2 + (y-3)^2 + (z-2)^2} = \sqrt{(x-1)^2 + y^2 + (z-4)^2} \\ &\Leftrightarrow (x+1)^2 + (y-3)^2 + (z-2)^2 = (x-1)^2 + y^2 + (z-4)^2 \Leftrightarrow 4x - 6y + 4z = 3. \end{aligned}$$

Therefore the coordinates x, y, z satisfy the equation: $\boxed{4x - 6y + 4z = 3}$.

- (d) Write down the equation of a sphere with centre (x_0, y_0, z_0) and radius r .

By definition, the sphere with centre (x_0, y_0, z_0) and radius r has equation:

$$\boxed{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = r^2}.$$

- (e) Find the equation of a sphere with one of its diameters having end-points $A(-3, 2, -5)$ and $B(1, 4, -1)$.

The centre of the sphere is the midpoint of the segment with end-points $A(-3, 2, -5)$ and $B(1, 4, -1)$. That is

$$\left(\frac{-3+1}{2}, \frac{2+4}{2}, \frac{-5-1}{2} \right) = (-1, 3, -3).$$

Its radius is

$$r = \frac{|\vec{AB}|}{2} = \frac{1}{2} \sqrt{(1+3)^2 + (4-2)^2 + (-1+5)^2} = \frac{\sqrt{36}}{2} = 3.$$

Therefore, the sphere has equation:

$$\boxed{(x+1)^2 + (y-3)^2 + (z+3)^2 = 9}.$$

(f) Find the centre and radius of the sphere whose equation is

$$x^2 + 6x + y^2 + 4y + z^2 - z + \frac{45}{4} = 0.$$

By completing the squares, we get

$$x^2 + 6x + y^2 + 4y + z^2 - z + \frac{45}{4} = 0 \Leftrightarrow \boxed{(x+3)^2 + (y+2)^2 + (z-\frac{1}{2})^2 = 2}$$

This is the sphere with centre $(-3, -2, 1/2)$ and radius $\sqrt{2}$.

2. The vector function $\vec{r}(t) = \left(\frac{1}{\arccos(t-2)}, \arctan(t+1), \ln|2t-3| \right)$ has domain $[1, \frac{3}{2}) \cup (\frac{3}{2}, 3)$.

3. (a) Since the limit of each component exists, we get that

$$\begin{aligned} \lim_{t \rightarrow 0^+} \left(\frac{1}{\pi} \arctan\left(\frac{1}{t}\right), -\sqrt{t} \sin\left(\frac{1}{t^2}\right), e^t \right) &= \left(\lim_{t \rightarrow 0^+} \frac{1}{\pi} \arctan\left(\frac{1}{t}\right), -\lim_{t \rightarrow 0^+} \sqrt{t} \sin\left(\frac{1}{t^2}\right), \lim_{t \rightarrow 0^+} e^t \right) \\ &= \left(\frac{1}{2}, 0, 1 \right). \end{aligned}$$

Use the squeeze theorem for $\lim_{t \rightarrow 0^+} \sqrt{t} \sin\left(\frac{1}{t^2}\right)$. In fact,

$$-1 \leq \sin\left(\frac{1}{t^2}\right) \leq 1 \quad \Rightarrow \quad -\sqrt{t} \leq \sqrt{t} \sin\left(\frac{1}{t^2}\right) \leq \sqrt{t}.$$

(b) Since the limit of each component exists, we have

$$\lim_{t \rightarrow 0^+} \left(\frac{\sin(2t)}{t}, t \ln t, \sqrt{t} \sin^2\left(\frac{1}{t^2}\right) \right) = \left(\lim_{t \rightarrow 0^+} \frac{\sin(2t)}{t}, \lim_{t \rightarrow 0^+} t \ln t, \lim_{t \rightarrow 0^+} \sqrt{t} \sin^2\left(\frac{1}{t^2}\right) \right) = (2, 0, 0).$$

Use squeeze theorem for $\lim_{t \rightarrow 0^+} \sqrt{t} \sin^2\left(\frac{1}{t^2}\right)$.

4. For each of the following, sketch the curve in $\mathbb{R}^2$ parameterized by:

- (i) $\vec{r}(t) = (\cos t, \sin t), \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}.$
- (ii) $\vec{r}(t) = \left(\sqrt{1 - (\ln t)^2}, \ln t \right), \quad e^{-1} \leq t \leq e.$

First write down the cartesian equations in order to identify the shape of each curve involved.

What do you observe? The two vector functions represent the semi-circle with center $(0, 0)$ and radius 1, lying in the semi-plane $x \geq 0$.

5. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}(t) := \left(\frac{1-t^2}{1+t^2}, e^t, \frac{2t}{1+t^2}\right)$.

(i) Say whether the curve $\mathcal{C}$ lies on the cylinder $x^2 + z^2 = 1$.

Setting $x = \frac{1-t^2}{1+t^2}$ and $z = \frac{2t}{1+t^2}$, we find that

$$x^2 + z^2 = \left(\frac{1-t^2}{1+t^2}\right)^2 + \left(\frac{2t}{1+t^2}\right)^2 = \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{(1+t^2)^2}{(1+t^2)^2} = 1.$$

(ii) Evaluate the indefinite integral $\int \vec{r}(t) dt$.

$$\begin{aligned} \int \vec{r}(t) dt &= \left(\int \frac{1-t^2}{1+t^2} dt, \int e^t dt, \int \frac{2t}{1+t^2} dt \right) \\ &= \left(\int \left(\frac{2}{1+t^2} - 1 \right) dt, \int e^t dt, \int \frac{2t}{1+t^2} dt \right) \\ &= (2 \arctan t - t + c_1, e^t + c_2, \ln(1+t^2) + c_3). \end{aligned}$$

6. We seek t such that $2(t^2 + 3) - (2t^2) - (t - 1) = 3$. That is, $t = 4$ which corresponds to the point $(19, 32, 3)$.

7. Consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [0, \frac{3\pi}{4}] \rightarrow \mathbb{R}^2$ defined by

$$\vec{r}(t) = (3 \sin(2t), 2 \cos(2t)) \quad \text{for every } 0 \leq t \leq \frac{3\pi}{4}.$$

Sketch the curve $\mathcal{C}$ indicating the points of intersection with the axes.

Write down the parametric equations: $x = 3 \sin(2t)$, $y = 2 \cos(2t)$ with $0 \leq t \leq \frac{3\pi}{4}$. We eliminate the parameter t in the parametric equations by using the trig. identity $\sin^2(2t) + \cos^2(2t) = 1$ to get the equation of the ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$. This means that the $\mathcal{C}$ lies on the ellipse. Now, since $0 \leq 2t \leq 3\pi/2$, we get that the curve $\mathcal{C}$ is the arc of ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ lying in the first, third and fourth quadrants.

8. Evaluate:

(a) $\int_0^{\sqrt{\frac{\pi}{2}}} (t \sin(t^2), 2t^3, \sqrt{2}) dt$. Following a componentwise operation, we find that

$$\begin{aligned} \int_0^{\sqrt{\frac{\pi}{2}}} (t \sin(t^2), 2t^3, \sqrt{2}) dt &= \left(\int_0^{\sqrt{\frac{\pi}{2}}} t \sin(t^2) dt, \int_0^{\sqrt{\frac{\pi}{2}}} 2t^3 dt, \int_0^{\sqrt{\frac{\pi}{2}}} \sqrt{2} dt \right) \\ &= \left(\left[-\frac{1}{2} \cos(t^2) \right]_0^{\sqrt{\frac{\pi}{2}}}, \left[\frac{1}{2} t^4 \right]_0^{\sqrt{\frac{\pi}{2}}}, \sqrt{\pi} \right) = (1/2, \pi^2/8, \sqrt{\pi}). \end{aligned}$$

(b) $\int \frac{1}{1+t^2}(1, t, t^2)dt$. Similarly, we find that

$$\begin{aligned}\int \frac{1}{1+t^2}(1, t, t^2)dt &= \left(\int \frac{1}{1+t^2} dt, \int \frac{t}{1+t^2} dt, \int \frac{t^2}{1+t^2} dt \right) \\ &= \left(\arctan t + c_1, \frac{1}{2} \ln(1+t^2) + c_2, t - \arctan t + c_3 \right) .\end{aligned}$$

(c) $\int_0^{\frac{\pi}{4}} (\sec t, \tan t, t \cos t)dt$. We find that

$$\begin{aligned}\int_0^{\frac{\pi}{4}} (\sec t, \tan t, t \cos t)dt &= \left(\int_0^{\frac{\pi}{4}} \sec t dt, \int_0^{\frac{\pi}{4}} \tan t dt, \int_0^{\frac{\pi}{4}} t \cos t dt \right) \\ &= \left(\int_0^{\frac{\pi}{4}} \frac{\sec t(\tan t + \sec t)}{\sec t + \tan t} dt, \int_0^{\frac{\pi}{4}} \frac{\sin t}{\cos t} dt, \int_0^{\frac{\pi}{4}} t \cos t dt \right) \\ &= \left(\left[\ln(\sec t + \tan t) \right]_0^{\frac{\pi}{4}}, \left[-\ln(\cos t) \right]_0^{\frac{\pi}{4}}, \left[t \sin t + \cos t \right]_0^{\frac{\pi}{4}} \right) \\ &= \left(\ln(\sqrt{2} + 1), \ln(\sqrt{2}), \frac{\sqrt{2}}{2} \left[\frac{\pi}{4} + 1 \right] - 1 \right) .\end{aligned}$$